

DIGITAL MUSIC STORE ANALYSIS

Advanced Business Intelligence & SQL Analytics Report

1. EXECUTIVE SUMMARY & KEY FINDINGS

This report details a comprehensive data-driven investigation into our global Digital Music Store. Utilizing structured database schema analytics, we unlocked vital insights concerning operational hierarchies, global consumer distributions, market optimization frontiers, and purchasing deep-dives. Our objective is to transition raw sales data into high-impact strategic business plans.

Core Findings Matrix:

- **Core Revenue Pillar:** The Rock music genre completely dominates consumer market baskets globally, representing the primary driving force behind customer retention and bulk track monetization.
- **Geographical Concentration:** USA, Canada, and Western Europe stand out as the highest volume generators, with specific metropolitan areas driving the vast majority of continuous operational revenue.
- **High-Value Spenders:** A minor percentage of VIP customers contribute to a disproportionate chunk of the total company revenue, showing an immediate need for an institutionalized Customer Lifetime Value (CLV) growth initiative.

2. BUSINESS INSIGHTS & STRATEGIC RECOMMENDATIONS

Targeted Experiential Marketing (Music Festivals)

Data indicates that localized regional capitals yield our highest aggregated invoice sales totals. Instead of spreading promotional budgets thin across full nations, capital should be focused exclusively on top-performing cities. We recommend organizing the promotional *Digital Music Store Festival* directly in the highest-earning city to maximize physical-to-digital cross-conversions.

Artist-Driven Lifecycle Monetization

The top 10 Rock bands generate a massive percentage of total track engagement. By utilizing detailed client spending metrics per artist, the marketing department can introduce hyper-targeted email campaigns. For instance, when an artist like the top-ranked rock band drops a new album, automated promotional tracks can be recommended directly to customers already flagged as high-tier spenders for that specific artist.

Global Genre Diversification & Localization

While Rock is the overall leader, secondary genres vary drastically per country border. In regions where multiple genres share the top purchase counts, marketing content must remain dynamically localized rather than monocultural. Standardizing global homepages hurts conversions; dynamic dashboard curation based on localized top ranks is essential.

3. TECHNICAL SQL SOLUTIONS & IMPLEMENTATIONS

Question Set 1: Easy Solutions

Q1: Most Senior Employee based on Job Title

Identifies organizational hierarchy structures to determine line-of-command ownership.

```
SELECT title, last_name, first_name
FROM employee
ORDER BY levels DESC
LIMIT 1;
```

Q2: Countries with the Most Invoices

Maps transactional volume frequency by geography to highlight absolute market penetration.

```
SELECT billing_country, COUNT(*) AS invoice_count
FROM invoice
GROUP BY billing_country
ORDER BY invoice_count DESC;
```

Q3: Top 3 Values of Total Invoice

Discovers the single highest transaction values historically logged inside our checkout systems.

```
SELECT DISTINCT total
FROM invoice
ORDER BY total DESC
LIMIT 3;
```

Q4: Best City for Promotional Music Festival

Aggregates total absolute invoice revenue grouped explicitly by city to find our primary revenue center.

```
SELECT billing_city, SUM(total) AS invoice_total
FROM invoice
GROUP BY billing_city
ORDER BY invoice_total DESC
LIMIT 1;
```

Q5: Top Spending Customer

Employs an **INNER JOIN** between customer profiles and transactional data to isolate the single highest contributing consumer.

```
SELECT c.customer_id, c.first_name, c.last_name, SUM(i.total) AS total_spent
FROM customer c
INNER JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id
ORDER BY total_spent DESC
LIMIT 1;
```

4. ADVANCED MULTI-JOIN ANALYTICS

Question Set 2: Moderate Solutions

Q1: Alphabetical Rock Music Listeners

Flattens a multi-tier normalized schema (Customer → Invoice → Line Item → Track → Genre) to discover target lists for genre-specific marketing campaigns.

```
SELECT DISTINCT c.email, c.first_name, c.last_name, g.name AS genre_name
FROM customer c
INNER JOIN invoice i ON c.customer_id = i.customer_id
INNER JOIN invoice_line il ON i.invoice_id = il.invoice_id
INNER JOIN track t ON il.track_id = t.track_id
INNER JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email ASC;
```

Q2: Top 10 Rock Artists by Track Volume

Aggregates individual track metadata across catalogs to discover which artists hold the largest market share within our core genre.

```
SELECT art.artist_id, art.name, COUNT(t.track_id) AS total_songs
FROM artist art
INNER JOIN album alb ON art.artist_id = alb.artist_id
INNER JOIN track t ON alb.album_id = t.album_id
INNER JOIN genre g ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY art.artist_id
ORDER BY total_songs DESC
LIMIT 10;
```

Q3: Tracks Exceeding Average Duration

Utilizes an independent scalar subquery to isolate tracks longer than the dynamic database mean: *Length* > μ .

```

SELECT name, milliseconds
FROM track
WHERE milliseconds > (
    SELECT AVG(milliseconds)
    FROM track
)
ORDER BY milliseconds DESC;

```

5. COMPLEX ANALYTICAL WINDOW FUNCTIONS

Question Set 3: Advance Solutions

Q1: Total Accumulative Spend per Customer per Artist

Evaluates granular product-level itemization metrics by computing $Total = \sum(UnitPrice \times Quantity)$ across cross-joined consumer profiles.

```

SELECT
    c.first_name || ' ' || c.last_name AS customer_name,
    art.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spent
FROM customer c
INNER JOIN invoice i ON c.customer_id = i.customer_id
INNER JOIN invoice_line il ON i.invoice_id = il.invoice_id
INNER JOIN track t ON il.track_id = t.track_id
INNER JOIN album alb ON t.album_id = alb.album_id
INNER JOIN artist art ON alb.artist_id = art.artist_id
GROUP BY c.customer_id, art.artist_id
ORDER BY total_spent DESC;

```

Q2: Most Popular Genre by Country (Handling Equal Ties)

Utilizes Common Table Expressions (CTEs) paired with `DENSE_RANK()` partitioned over geographical entities to accurately surface overlapping popularity scores.

```

WITH popular_genre AS (
    SELECT
        c.country,
        g.name AS genre_name,
        COUNT(il.quantity) AS purchases,
        DENSE_RANK() OVER(PARTITION BY c.country ORDER BY COUNT(il.quantity) DESC) AS rank_num
    FROM customer c
    INNER JOIN invoice i ON c.customer_id = i.customer_id
    INNER JOIN invoice_line il ON i.invoice_id = il.invoice_id
    INNER JOIN track t ON il.track_id = t.track_id
    INNER JOIN genre g ON t.genre_id = g.genre_id
    GROUP BY c.country, g.genre_id
)
SELECT country, genre_name, purchases
FROM popular_genre
WHERE rank_num = 1;

```

Q3: Top VIP Spenders per Country Border

Isolates high-net-worth customers across separate target international areas. The query accounts for shared first-place spenders via robust window partitioning.

```

WITH customer_spending_by_country AS (
    SELECT
        c.customer_id, c.first_name, c.last_name,
        i.billing_country AS country,
        SUM(i.total) AS total_spent,
        DENSE_RANK() OVER(PARTITION BY i.billing_country ORDER BY SUM(i.total) DESC) AS rank_num
    FROM customer c
    INNER JOIN invoice i ON c.customer_id = i.customer_id
    GROUP BY c.customer_id, i.billing_country
)
SELECT country, first_name, last_name, total_spent
FROM customer_spending_by_country
WHERE rank_num = 1
ORDER BY country ASC;

```